Implementation of Data Pipeline and ETL on Retail Store Dataset Using Pentaho

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Abstract*—This paper presents the implementation of a data pipeline and ETL process on a retail store dataset using Pentaho Data Integration. The dataset, sourced from a CSV file containing 12,575 rows and 11 columns, was processed through four main stages: data ingestion, cleaning, dimensional modeling, and visualization. During the cleaning stage, issues such as null values, duplicate records, and numerical inconsistencies between price, quantity, and total_spent were resolved using Pentaho transformation steps including If Field Value is Null, Unique Rows, Calculator, and Filter Rows. The processed data was then structured into a star schema consisting of one fact table and five dimension tables, stored in a MySQL database called retail_db. Finally, the data was connected to Power BI to build an interactive dashboard. The results showed a total revenue of 1.55M from 11,971 validated transactions, with Patisserie as the top-earning category and an almost equal split between in-store and online purchases. This project demonstrates that a structured ETL pipeline using open-source tools can effectively transform raw transactional data into reliable, analysis-ready output for business decision-making.

***Keywords*—ETL, Pentaho Data Integration, data warehouse, star schema, Power BI, retail analytics**

I. Introduction

Data engineering has become an essential part of modern business operations, acting as the backbone that connects raw data sources to the people who actually use them, such as analysts, data scientists, and business decision makers. Without a proper data pipeline, raw data often remains scattered and unprocessed, making it difficult to extract any meaningful insight from it.

This project applies data engineering practices to a retail sales transaction dataset. The dataset, sourced from a CSV file named retail_store_sales.csv, contains 12,575 rows and 11 columns which includes columns such as Transaction ID, Customer ID, Category, Item, Price Per Unit, Quantity, Total Spent, Payment Method, Location, Transaction Date, and Discount Applied. Like most real-world datasets, it contains issues such as null values, duplicate records, and numerical inconsistencies that need to be resolved before the data can be used reliably.

To handle this, an ETL pipeline was built using Pentaho Data Integration. The pipeline covers the full flow from ingestion and cleaning to dimensional modeling and visualization. Processed data is structured into a star schema consisting of one fact table and five dimension tables, then connected to a Power BI dashboard for business analysis.

The goal of this project is to build a proper data pipeline, from cleaning the raw data all the way to producing a dashboard that can be used for business analysis.

II. Data Architecture and Pipeline

The data architecture in this project follows a linear flow, starting from the raw CSV file all the way to a Power BI dashboard. The pipeline is divided into four stages.

1. Data Source: The starting point of the pipeline, where data is extracted from the raw file retail_store_sales.csv.
2. Data Processing: The stage where data cleansing and transformation are carried out, including handling null values, removing duplicates, and fixing inconsistencies.
3. Data Storage: Processed data is loaded into a MySQL database called retail_db, structured into a star schema with one fact table and five dimension tables.
4. Visualization: The final stage, where the data is connected to Power BI to build an interactive dashboard for sales analysis.

The first stage is data ingestion. The raw file retail_store_sales.csv is read using Pentaho and loaded into a staging table called raw_sales_clean in the retail_db MySQL database.

The second stage is data processing. Once in staging, the data goes through a cleaning transformation that handles null values, removes duplicates, fixes type mismatches, and resolves numerical inconsistencies between price, quantity, and total_spent. The output is stored in a separate table called clean_sales.

The third stage is dimensional modeling. From clean_sales, five dimension tables are built: dim_customer, dim_date, dim_location, dim_payment, and dim_product each containing deduplicated values from their respective

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columns. A fact table, fact_transactions, is then created by joining clean_sales against all five dimension tables and replacing natural keys with surrogate integer keys.

The final stage is visualization. The fact_transactions table along with the dimension tables are connected to Power BI, where an interactive dashboard is built for sales analysis.

All transformations are orchestrated through a single Pentaho job file, retail_pipeline_job.kjb, which runs each transformation in sequence starting from cleaning, followed by each dimension table, and ending with the fact table.

III. Methodology

A. ETL Approach

This project adopts the ETL (Extract, Transform, Load) approach, wherein data is cleansed and structurally modified prior to its insertion into the database. This approach was selected due to the necessity for specific data adjustments, such as modifying transaction date formats. This contrasts with the ELT (Extract, Load, Transform) method, which is generally more suited for massive-scale data systems processed directly within cloud storage environments.

B. Software Selection

The selection of software is tailored to the data scale and business requirements to ensure operational efficiency. The ETL process is executed using Pentaho software to perform data extraction from the CSV, execute transformations, and facilitate data migration to the target database. The utilization of Pentaho automates the entire workflow orchestration, making it highly structured and efficient.

C. Data Quality and Reliability

Data Quality checks are conducted to guarantee that the data is accurate, valid, and devoid of null variables or duplications. This validation is critical for maintaining Data Reliability, ensuring that the resulting final dashboard remains stable and avoids presenting misleading analytical metrics for decision-making purposes.

D. Implementation Details

The pipeline was built using four transformation files that executed sequentially through a single job file, retail_pipeline_job.kjb. The first transformation, retail_pipeline.ktr, reads the source CSV file and loads into a staging table called raw_sales_clean in the retail_db database.

Once the data is in staging, cleaning_sales.ktr handles cleaning process. Column names are renamed to snake_case and data types are adjusted for example, Price Per Unit becomes price with type Number, Quantity becomes Integer, and Transaction Date becomes Date. Null values are then handled: the item column defaults to "Unknown" and discount_applied defaults to FALSE. Duplicates are removed using a Unique Rows step based on Transaction ID and Item. Inconsistencies were found between the price, quantity, and total_spent columns, two Calculator steps were added to recompute all three values so they remain consistent with each other. Rows that still contain null values in critical columns such as item, quantity, or total_spent are discarded via a Filter Rows step, and only valid rows are written to the clean_sales table.

The next five transformations dim_customer.ktr, dim_date.ktr, dim_location.ktr, dim_payment.ktr, and dim_product.ktr each pull data from clean_sales, deduplicate it, and load it into their respective dimension tables. Finally, fact_table.ktr performs DB Lookups against each dimension table to replace natural keys like customer_id and location with integer surrogate keys, and writes the result to the fact_transactions table.

IV. Data Modelling

Data Modelling is architected using a Star Schema approach to optimize query execution times, make it easier to query and read in BI tools. This model centralizes the primary transaction metrics into a single Fact table, which is surrounded by five supporting Dimension tables. The schema structure consists of:

- Fact Table: Serves as the central table storing all foreign keys (FK) alongside quantitative metrics such as *quantity, total spent,*and *discount applied*.
- DIM_CUSTOMER: A customer dimension table storing unique identities via customer_id and customer_code
- DIM_Product: An item detail table storing item details such as price, item name, and category.
- DIM_LOCATION: A table recording the codes, types, and names of transaction locations.
- DIM_PAYMENT: A table logging the payment methods and categories utilized by customers.
- DIM_DATE: A table decomposing transaction time (granularity) into elements of year, quarter, month, day, and day of the week.

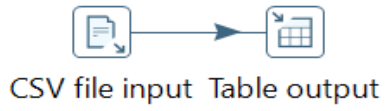


Fig. 1. Data ingestion flow in retail_pipeline.ktr.

Fig. 1 shows the data ingestion step in retail_pipeline.ktr. The process is straightforward the raw file retail_store_sales.csv is read using a CSV File Input step and immediately loaded into the staging table raw_sales_clean in the retail_db database. No transformation is applied at this stage; the data is taken as-is from the source file.

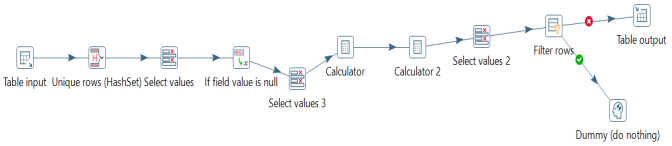


Fig. 2. Cleaning transformation in cleaning_sales.ktr.

Fig. 2 shows the complete cleaning flow in cleaning_sales.ktr. The process starts with a Table Input step

that reads from the staging table, followed by a Unique Rows (HashSet) step to remove duplicate records based on Transaction ID and Item. A Select Values step then renames columns to snake_case and enforces the correct data types for each field. The If Field Value is Null step handles missing values by assigning default values to columns like item and discount_applied. Two Calculator steps are used to resolve numerical inconsistencies price, quantity, and total_spent are recomputed so that all three values remain consistent with each other. A second Select Values step is applied after the calculation to finalize the column structure. Finally, a Filter Rows step separates valid rows from invalid ones, where rows with null values in critical columns are sent to a Dummy step and discarded, while clean rows are written to the clean_sales table via Table Output.

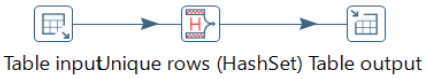


Fig. 3. Dimension table transformation flow.

Fig. 3 shows the transformation flow used to build each dimension table. The Table Input step selects only the relevant columns from clean_sales based on what each dimension needs for example, dim_customer only takes customer_id, while dim_product takes category, item, and price. A Unique Rows (HashSet) step then removes duplicate entries to ensure each dimension only contains distinct values. The deduplicated result is then loaded into the respective dimension table via Table Output, where a surrogate primary key is automatically assigned. This same flow is applied separately for all five dimension tables, with the only difference being the columns selected at the Table Input stage.



Fig. 4. Dimension and fact table pipeline in Pentaho.

Fig. 4 shows the fact table transformation flow in fact_table.ktr. Starting from a Table Input that reads all relevant columns from clean_sales, the data passes through five sequential DB Lookup steps one for each dimension table (dim_customer, dim_product, dim_location, dim_payment, and dim_date). Each lookup replaces the original natural key with the corresponding surrogate integer key generated from the dimension table. The final output is then loaded into the fact_transactions table, which serves as the central table in the star schema.



Fig. 5. Job orchestration in retail_pipeline_job.kjb.

Fig. 5 shows the full job orchestration in retail_pipeline_job.kjb. The job starts with a Start entry point and executes each transformation in a fixed sequence cleaning_sales first, then dim_customer, dim_product, dim_date, dim_location, dim_payment, and finally fact_table. Each step is connected by a green line indicating successful execution before moving to the next transformation. The job ends at a Success checkpoint, confirming that all transformations have completed without errors.

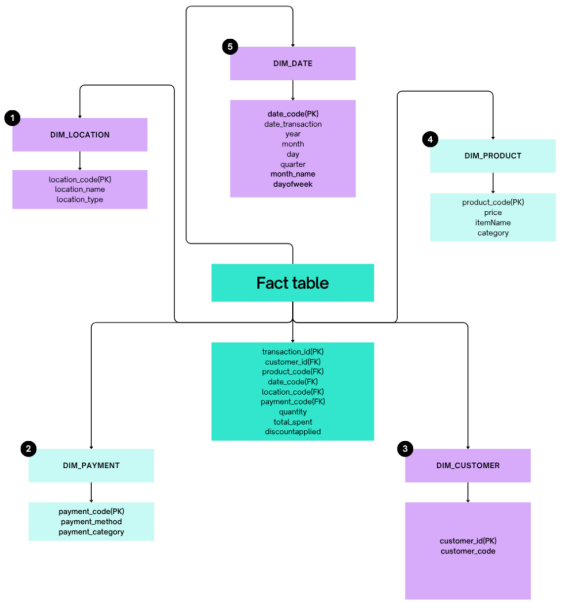


Fig. 6. Star schema diagram of retail_db.

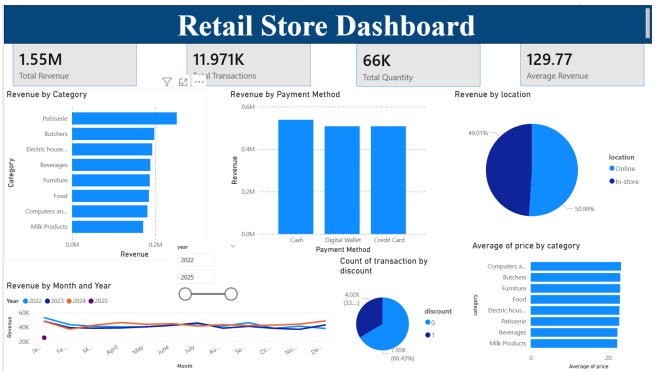
Fig. 6 shows the star schema structure of retail_db. The fact table sits at the center and stores all foreign keys customer_id, date_code, product_code, location_code, and payment_code alongside the quantitative metrics quantity, total_spent, and discount_applied. Surrounding the fact table are five dimension tables: DIM_LOCATION which stores location_code, location_name, and location_type; DIM_DATE which breaks down the transaction date into year, month, day, quarter, month_name, and dayofweek; DIM_PRODUCT which holds product_code, price, itemName, and category; DIM_PAYMENT which records payment_code, payment_method, and payment_category; and DIM_CUSTOMER which stores customer_id and customer_code.

V. Results and Dashboard Visualization

The data that progressed through the orchestration and modeling stages was successfully visualized into a centralized Retail Store Dashboard. Based on the processed, consumption-ready data, key performance metrics indicate a

Total Revenue reaching 1.55M, derived from a record of 11.971K validated transactions.

In terms of product contribution, revenue by category reveals that Patisserie products dominate top earnings, followed by the Butchers segment, electric housewares, and Beverages. Demographic location analysis demonstrates a highly competitive transaction distribution, with physical (In-store) purchases marginally leading at a 50.99% share, while Online orders account for 49.01%. Within the payment method segment (revenue by payment method), the monetary distribution among Cash, Digital Wallet, and Credit Card transactions exhibits remarkably close trends, all hovering in the range of 0.4M to nearly 0.6M.



VI. Conclusion

The goal of this project was to build a data pipeline that could take raw retail transaction data and turn it into something usable for business analysis. We Start from a CSV file with 12,575 rows, the pipeline went through ingestion, cleaning, dimensional modeling, and visualization before the data was ready to use.

During the cleaning stage, several issues were found in the raw data. Some rows had null values in critical columns like item and discount_applied, others were duplicates, and there were numerical inconsistencies where price, quantity, and total_spent did not match each other. All of these were handled in cleaning_sales.ktr before any data passed to the dimension tables.

The star schema model made it easier to work with the data in Power BI. Instead of querying one large flat table, the fact table could be joined to any dimension tables depending on what needed to be analyzed. This kept the queries simple and the dashboard responsive.

From the dashboard, total revenue came out to 1.55M from 11,971 transactions, with an average revenue of 129.77 per transaction and 66K total units sold. Patisserie was the highest earning category, followed by Butchers, Electric Housewares, and Beverages. The split between in-store and online purchases was almost even at 50.99% and 49.01%, and the three payment methods Cash, Digital Wallet, and Credit Card were all in a similar range between 0.4M and 0.6M.

Overall, the pipeline worked as intended. The data that came out was clean, the model was structured properly, and the dashboard gave a clear picture of the retail store's sales performance.

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